

Question Paper for 1st Periodical Examination (March 2026) (Session Jan-May 2026)

Time: One and a half hours
 Note: Attempt all questions

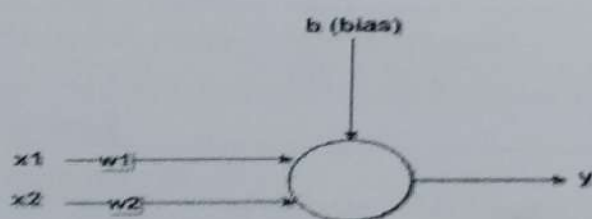
Max Marks:30

CO	COURSE OUTCOMES
CO1	Understand fundamentals concepts, models of deep learning.
CO2	Apply deep learning algorithms to learn a model.
CO3	Implement and analyze various tools to learn efficiently like variants of gradient descent.
CO4	Apply autoencoders in various applications
CO5	Apply CNN, RNN and other models to learn from various types of datasets like image, text data set.

QNo.	Questions	Marks	CO	KL
1.	Do the following:			
1.a	How learning rates in neural networks are decided?	1	1	L1
1.b	Explain the exploding gradient problem?	1	3	L1
1.c	Find out the stationary points of the function $f(x) = 3x^2 + 9x + 12$	1	1	L2
1.d	What is a sigmoid function?	1	1	L1
1.e	The number of learnable parameters in a fully connected neural network with architecture (input_layer_size=12, first_hidden_layer_neurons=8, second_hidden_layer_neurons=3, output_layer_size=2) are	1	1	L1
2.	Compare and critically examine gradient descent, Momentum based GD and Nesterov Accelerated GD.	5	3	L2
3.	Show that the derivative of sigmoid function $f(x)$ is $f'(x) = f(x)(1-f(x))$. Plot a sigmoid function and its derivative?	5	1	L3
4.	Explain back propagation algorithm in detail.	5	3	L3
5.	Compare batch, stochastic and mini-batch gradient descent approaches to update new weights. Explain their disadvantages and advantages.	10	2	L4

OR

Consider the neural network and dataset below for the regression problem. Learn weights using batch GD method and stochastic GD(1 iteration each). Assume $w_1=1.5$, $w_2=1.5$, $b=-6$, learning rate= 0.1)



x1	x2	y
5	6	10
2	-7	-19

CO: Course Outcome

KL: Knowledge Level: L1 – Remember L2 – Understand L3 – Apply L4 Analyze L5 – Evaluate L6 – Create

Course Advisor: Rahul Pal Singh

Designation: Assistant Professor